

Victor Pavao

GIS Analyst

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SKILLS

- **Languages:** Python, JavaScript/TypeScript, SQL (Spatial SQL), R, Arcade, HTML/CSS, Java, Git
- **Libraries/Frameworks:** Postgres/PostGIS, Carto, Geopandas, Node.js, React.js, AWS, Google Firebase
- **Geospatial Software:** ArcGIS Pro, ArcGIS Online, ArcGIS Field Maps, Survey123, QGIS, ArcGIS Enterprise, pgAdmin, GDAL, FME, GitHub/GitLab, OpenStreetMap (OSM)

EXPERIENCE

Connecticut Transportation Institute, Storrs, CT – GIS Research Technician

SEPTEMBER 2025 - PRESENT

- Utilizing ArcGIS Pro to create and edit geometric features that match pavement markings on state highways
- Publishing new feature layers using ArcServer and ArcGIS Enterprise
- Pavement markings database to be used by Connecticut Department of Transportation to analyze car crashes and estimate project costs

University of Massachusetts, Amherst, MA – Graduate Teaching Assistant

SEPTEMBER 2025 - PRESENT

- Assisting professor in developing course materials for graduate-level GIS Programming course, which covers Python, arcpy, Jupyter Notebooks, Model Builder and related topics
- Completing weekly grading tasks and communicating feedback to students

University of Massachusetts, Amherst, MA – Research Assistant (GIS Specialist)

MAY 2025 - PRESENT

- Authoring Python scripts that estimate private property well usage in Massachusetts as part of larger Source Water Protection and Management Project for Massachusetts DEP Well Driller Program
- Creating and administering ArcGIS Online dashboard to display relevant data for stakeholders
- Performing QA on output layers using ArcGIS Field Maps
- Providing input on supervisor's Survey123 project for collecting data about wells
- Researching and documenting well driller regulations in all 50 states for Massachusetts DEP Well Driller Program

Kestrel Land Trust, Amherst, MA – Land Stewardship Coordinator

AUGUST 2024 - MAY 2025

- Acted as main GIS resource at organization, significantly increasing conservation organization's GIS capabilities
- Created trail maps and property assessments with ArcGIS Pro; work was featured on grant applications, trail kiosks, and internal assessments
- Utilized ArcGIS geoprocessing tools for georeferencing, geocoding, converting file types, and buffering
- Reorganized and managed all GIS data on common file server
- Developed Python toolboxes and Jupyter Notebook scripts for reusable workflows; one joins parcel shapefiles and assessors' tables across municipalities and produces a single output shapefile
- Uploaded missing trail data to OpenStreetMap; downloaded OpenStreetMap data using Overpass API and GDAL tools
- Mapped trails on the ground with GPS unit and assisted in trail maintenance work
- Trained other staff in GIS workflows and file storage practices

SAVVI Financial, Waltham, MA – Software Engineer

JANUARY 2022 - NOVEMBER 2023

- Built several major components of financial services application using React/TypeScript/MaterialUI
- Collaborated with back-end team to display custom front-end content for company partners
- Led implementation of end-to-end and unit testing using Playwright
- Assisted with re-skinning and migrating app from React Native to React
- Submitted work in a PR-driven CI/CD workflow

Coresecure, Cambridge, MA – Software Engineer

APRIL 2020 - DECEMBER 2021

- Created web applications for pharmaceutical clients
- Front-end libraries used for applications include Nuxt (Vue), React, and jQuery
- Implemented Contentful CMS in several client projects
- Integrated end-to-end testing project using Cypress

Goodpath, Cambridge, MA – Software Engineer

SEPTEMBER 2019 - MARCH 2020

- Helped create web application for healthcare startup in its early stages using React, Gatsby, and Google Firebase
- Implemented front-end payment system with Stripe library
- Developed company's custom style sheets with SCSS

EDUCATION

University of Massachusetts, Amherst – M.S. Geographic Information Systems Technology

SEPTEMBER 2024 - DEC 2025

Villanova University – B.S. Computer Science

AUGUST 2014 - MAY 2018

CERTIFICATIONS

Integrating Data with FME Platform (Safe Software)

APRIL 2025

PROJECTS

Geospatial Platform Project

- Developed a full-stack web application for hosting and displaying geospatial data, mentored by Spatial Databases professor
- Project implemented with podman for containerizing PostGIS database, feature services, API layer and front-end application, with the goal of hosting containers on Red Hat OpenShift on AWS

Tournament Management System

- Created serverless web application to manage trivia tournaments for quiz leagues as part of long-term contract with quiz organization
- Implemented project with React and Typescript front-end, Google Firebase database, and Azure App Services for hosting

AWARDS

2026 AAG Geobowl – Most Valuable Player

- Finished first in individual points at the 2026 American Association of Geographers (AAG) conference Geography Bowl